

JOSHUA COWELL

Active Top Secret / SCI Clearance | Full-Scope Polygraph

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected May 2027

Liberty University, Lynchburg, VA

GPA: 3.85

Selected Coursework: Dynamics, Fluid Dynamics, Heat Transfer, Thermodynamics I & II, Differential Equations, Thermal-Fluids Laboratory

Founding Member: American Nuclear Society (ANS) student chapter (2025); AIAA student chapter (2026)

Leadership: Liberty Rocketry – Chairman, Board of Advisors (Jul 2026 – Present)

ENGINEERING EXPERIENCE

Advanced Weapons Systems Analyst Intern

May 2026 – Present

Central Intelligence Agency (CIA)

- Conduct testing of warhead systems and analyze results to calibrate simulation models
- Develop a Python-based geospatial analysis tool for operational planning that meets team requirements
- Conduct technical assessments of advanced unmanned systems with multidisciplinary engineering teams, translating findings into actionable intelligence for senior leadership and policymakers
- Brief senior Agency officials and write concise analytic products that communicate complex technical data in decision-ready formats

Thermal Hydraulics Engineering Intern

May 2025 – Dec 2025

Framatome Nuclear

- Developed and validated a Python predictive-wear model against documented component wear, reducing prediction error by 75% versus the legacy model
- Built a Python/xarray algorithm processing six-dimensional datasets to calculate safe operating limits, reducing analysis time from 800+ hours to minutes
- Performed system-level thermal-hydraulic and flow-induced-vibration (FIV) analyses using AFT Fathom and custom tools to support component qualification
- Authored technical documentation and presented findings to engineering leadership; received a \$2,500 technical-presentation award

Chief Systems Engineer and Team Lead

Jun 2025 – Jul 2026

Liberty Rocketry

Earlier roles: Assistant Chief Engineer; Senior Structural Engineer; Fin Fluid-Dynamics Engineer

- Placed 18th of 141 international collegiate teams at the 2026 IREC
- Led a 60-member student team in the design, testing, and launch readiness of a high-powered rocket
- Oversaw engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Facilitated communication among internal teams, external vendors, and university faculty to improve project execution
- Reviewed SolidWorks, ANSYS, MATLAB, OpenRocket, and RocketPy models and simulations used for team design decisions

Lead Undergraduate Research Assistant

Sep 2024 – Present

Liberty University

- Lead a five-person team conducting split-Hopkinson pressure bar (SHPB) testing for dynamic material characterization
- Develop test procedures and fixtures; use PicoScope strain data to generate stress-strain curves and correlate high-speed imaging with SEM results

TECHNICAL SKILLS

Systems & Test:	System integration, test planning, model calibration, subsystem coordination, technical documentation
Thermal & CAE:	Thermal-hydraulics, heat transfer, FIV, ANSYS Fluent, ANSYS Mechanical, AFT Fathom, CFD, FEA
CAD & Manufacturing:	SolidWorks, Onshape, GD&T, technical drawings, 3D printing, CNC machining
Programming & Lab:	Python (NumPy, pandas, xarray), MATLAB, SHPB testing, SEM, high-speed imaging

SELECTED PROJECTS

Heat Exchanger Thermal-Hydraulic Optimization – MATLAB, ANSYS Fluent, SolidWorks

- Wrote MATLAB optimization code that evaluated 104 million crossflow and 287 million annular geometries against thermal, dimensional, and annual-cost constraints
- Validated the selected annular design against ANSYS Fluent CFD across heat-transfer and effectiveness metrics

Unmanned Aerial Vehicle Design & Analysis – SolidWorks, ANSYS Fluent, 3D Printing

- Designed and integrated a modular UAV airframe, propulsion system, and payload using replaceable 3D-printed components; evaluated performance with CFD